



Internship Daily Report

Day 39 | Spectra Systems | CloudLabs Division

Date: 26 March 2026 **Division:** Cloud Lab Engineering — CloudLabs Product Team



Client Served

Client Name: Contoso (Hackathon Event Client) **Project Type:** Cloud Hackathon Lab Environment Setup **Platform Used:** CloudLabs Admin Portal (admin.cloudlabs.ai) **Cloud Provider:** Microsoft Azure



Project Overview

The client — **Contoso** — approached Spectra Systems to design, configure, and deliver a fully managed **weekend hackathon lab environment** for 15 teams of 3–4 participants each. The environment was hosted on the **CloudLabs platform**, powered by **Microsoft Azure**, and required precise configurations covering team collaboration, automated resource provisioning, branding, and participant experience.

My role today was to understand the client requirements in full detail, map each requirement to the correct CloudLabs configuration, and produce and deploy all deliverables end to end.



Client Requirements — In Detail

Requirement 1 — Team-Based Shared Environment

What the client needed: Teams of 3–4 participants must collaborate inside a single shared environment. All team members must be able to see and manage resources deployed by their teammates.

How it was addressed:

- Subscription Type set to **Dedicated Subscription** — one full Azure subscription assigned per team
 - All team members granted **Contributor role at Subscription scope** via CloudLabs Template Permissions
 - Shared Subscription was evaluated and ruled out as it only gives Resource Group level access, which would isolate teammates from each other
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Requirement 2 — Automatic Service Principal Creation

What the client needed: The hackathon involves writing scripts that require a Service Principal for authentication. The Service Principal must be created automatically.

How it was addressed:

- **Create Service Principal** checkbox enabled in the CloudLabs Template
 - **Send Service Principal** checkbox enabled — Application ID, Client Secret, Tenant ID, and Subscription ID are automatically surfaced on the participant's lab details page
 - Participants use these credentials directly in Azure CLI commands to authenticate non-interactively
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Requirement 3 — Windows 11 VM via ARM Template

What the client needed: The ARM template must automatically deploy a Virtual Machine running Windows 11 when the lab launches.

How it was addressed:

- ARM template designed with Windows 11 image reference and deployed via the CloudLabs **ARM Template tab**
 - ARM template file and parameter file uploaded to Azure Blob Storage with public URLs and mapped to the template
 - **Enable VM Access Over HTTP** enabled — participants access the VM directly from their browser
 - **Virtual Machine Configuration** tab set up with VM name, DNS, admin username and password from ARM template outputs
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Requirement 4 — VM Duration Limit at Half the Lab Duration

What the client needed: The VM must have a usage limit of up to half of whatever the lab duration is set to.

How it was addressed:

- **VM Uptime Limit** in the ODL set to exactly **60 minutes** — 50% of the 120-minute total lab duration
 - Once the VM uptime limit is reached, the VM automatically shuts down
 - This enforces cost control while giving teams sufficient time to complete CLI-heavy tasks
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Requirement 5 — Sponsor Logo on Registration Page

What the client needed: The lab registration page must display the hackathon sponsor's logo for branding purposes.

How it was addressed:

- **Custom Logo URL** populated in the CloudLabs Template with the sponsor logo image URL
 - **Custom ODL Page Logo URL** also set at the ODL level
 - **Custom Page Title** set to **Contoso Hackathon 2024** at both Template and ODL levels
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Requirement 6 — Lab Preview Before Registration

What the client needed: Participants must be able to preview the lab before registering to evaluate whether the environment meets their team's needs.

How it was addressed:

- **Enable Lab Preview** checkbox enabled at both Template and ODL levels
 - A 2-page **Lab Preview JSON file** created covering hackathon overview, prizes, environment details, rules, and judging criteria
 - Preview JSON hosted on Azure Blob Storage and URL provided in the **Preview URL** field
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Requirement 7 — 15 Teams, One Attempt Each

What the client needed: The hackathon has exactly 15 teams and each team gets only one attempt — no resets or redeployments.

How it was addressed:

- **Maximum number of Users** set to **17** (15 teams + 2 buffer)
 - **Maximum Retake Count** set to **0** — no re-registration allowed
 - **Enable Hackathon** checkbox enabled in the ODL
 - Subscription Group configured with 15 Dedicated Subscriptions
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CloudLabs Template Created

The following configuration was completed inside the **CloudLabs Admin Portal** → **Templates** section:

Field	Value Configured
Template Name	Contoso-Hackathon-2024
Cloud Platform	Microsoft Azure
Cloud Usage Type	Cloud Resource Usage
Template Code	contosohack
Subscription Type	Dedicated Subscription
Custom Page Title	Contoso Hackathon 2024
Custom Logo URL	Sponsor logo URL configured
Owner Email	Spectra Systems project lead email
Deployment Plan	Resource Group — One
Region	East US
Lab Guide URL	GitHub master document URL configured
Approx. Deployment Duration	15 Minutes
ARM Template URL	Azure Blob Storage URL — contoso-hackathon.json
Parameter File URL	Azure Blob Storage URL — contoso-hackathon.parameters.json
Create Service Principal	 Enabled

Send Service Principal	 Enabled
Enable VM Access Over HTTP	 Enabled
Enable Lab Preview	 Enabled
Preview URL	Azure Blob Storage URL — contoso-hackathon-preview.json
Show Resources Tab	 Enabled
Github Master Document URL	Azure Blob Storage URL — v2-master-doc.json

Template Permissions configured:

Permission Type	Profile	Scope Level	Permission	Launch Type
Azure Built-in Role	Attendee	Subscription	Contributor	Apply at Launch

Virtual Machine Configuration set up:

Field	Value
VM Name	contoso-vm
Type	RDP
Server DNS Name	ARM template output parameter
Admin Username	ARM template output parameter
Admin Password	ARM template output parameter
Is Default VM	 Yes

ODL Created Using CloudLabs

The following configuration was completed inside the **CloudLabs Admin Portal** → **On Demand Labs** section:

Field	Value Configured
ODL Name	Contoso-Hackathon-Weekend-Mar2026
Template Mapped	Contoso-Hackathon-2024
Region	East US
Status	Registration Open
Approval Method	Registration Required
Duration	120 Minutes
VM Uptime Limit	60 Minutes (50% of Duration)
Expiry Date	Hackathon end date configured
Maximum Number of Users	17 (15 teams + 2 buffer)
Maximum Retake Count	0 (one attempt per team)
Subscription Group	15 Dedicated Subscriptions configured
Owner Email	Spectra Systems project lead email
Custom ODL Page Title	Contoso Hackathon 2024
Custom ODL Page Logo URL	Sponsor logo URL configured
Lab Guide URL	GitHub master document URL configured
Github Master Document URL	Azure Blob Storage URL — v2-master-doc.json
Lab Launch Page Description	Sponsor-branded prize and rules content configured
Enable Hackathon	☒ Enabled
Enable Lab Preview	☒ Enabled
Preview URL	Azure Blob Storage URL — contoso-hackathon-preview.json
Enable Uptime Alert	☒ Enabled

Enable Resource Operations  Enabled

Is Trial Lab  Not marked — Production environment

Deliverables Completed Today

#	Deliverable	Status
1	CloudLabs Template created and fully configured	 Done
2	ODL created, mapped to template, and configured	 Done
3	ARM Template built and uploaded to GitHub	 Done
4	Lab Preview JSON file created and hosted	 Done
5	Lab Launch Page Description written and configured	 Done
6	Template and ODL Description written and configured	 Done
7	Lab Guide — Introduction + Problem Statement	 Done
8	Lab Guide — Exercise 1 (Authentication via SP)	 Done
9	Lab Guide — Exercise 2 (VM Deployment + Verification)	 Done
10	Master Document JSON created and linked in CloudLabs	 Done
11	All lab guide files pushed to GitHub (syedfuzailabdulla/Contoso)	 Done

Key Learnings Today

#	Learning
1	Understood the difference between Shared and Dedicated Subscription in CloudLabs and when each applies
2	Learned how Service Principals are auto-created in CloudLabs and used for non-interactive Azure CLI authentication

- 3 Understood how VM Uptime Limit dynamically relates to the total lab Duration — always 50%
 - 4 Learned the Lab Preview JSON structure, the 2-page limit, and the fallback priority chain
 - 5 Understood how Enable Hackathon and Maximum Retake Count = 0 enforce team structure and single-attempt policy
 - 6 Understood Deployment Plan and chose Resource Group One for beginner participants
 - 7 Practised writing Azure CLI commands for VM deployment including VNet, Subnet, Public IP, NIC, and az vm create
 - 8 Learned how CloudLabs Master Document JSON links multiple GitHub markdown pages into a structured lab guide
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Day Timeline

Time	Activity
Session Start	Received and analysed the Contoso Hackathon client requirement
Phase 1	Broke down all 7 client requirements and mapped each to CloudLabs features
Phase 2	Determined correct subscription type with full justification
Phase 3	Built ARM Template for Windows 11 VM with auto Service Principal
Phase 4	Created CloudLabs Template with all configurations and permissions
Phase 5	Created ODL mapped to the template with all hackathon settings
Phase 6	Produced Lab Preview JSON, Launch Page Description, and Template Description
Phase 7	Wrote complete 3-page Lab Guide with problem statement, CLI exercises, and submission checklist
Phase 8	Created Master Document JSON and pushed all files to GitHub
Session End	All deliverables completed, configured, and live on CloudLabs

Day 39 Summary

Today was a **complete end-to-end client delivery day**. The entire Contoso Hackathon lab environment was designed, built, configured, and deployed — from the CloudLabs Template and ODL setup to the ARM template, lab guide, preview file, and GitHub publication. All 11 deliverables were completed and the environment is fully live and ready for the 15 hackathon teams to register and launch.
